

## MAJOR PROJECT REGISTRATION FORM

**Date:**

|                    |                                            |                                                 |           |      |
|--------------------|--------------------------------------------|-------------------------------------------------|-----------|------|
| Project Category:  | <input type="checkbox"/> Research oriented | <input type="checkbox"/> Application oriented   | Batch ID: | C-14 |
|                    | <input type="checkbox"/> Product oriented  | <input type="checkbox"/> Other (please specify) |           |      |
| Specify if others: |                                            |                                                 |           |      |

**Title of the Project:** AI-Powered Interview Preparation System with Personalized Ai Tutor

### Abstract

Campus placements and technical recruitment have become highly competitive, making interview preparation an important part of every student's career. However, many students and fresh graduates face difficulties in interviews due to a lack of proper guidance, limited practice, and low confidence. Most existing interview preparation platforms provide mock interviews and basic feedback, but they do not offer personalized learning or explain the mistakes made by the candidate. As a result, students often repeat the same mistakes without improving their overall interview skills.

The proposed AI-Powered Interview Preparation System with Personalized AI Tutor is designed to overcome these limitations by combining interview practice with continuous learning. The system is mainly developed for undergraduate students, fresh graduates, and job seekers preparing for campus placements. It conducts role-based and resume-based mock interviews using Artificial Intelligence (AI) and Natural Language Processing (NLP). After each interview, the system evaluates the candidate's responses and provides detailed feedback based on technical knowledge, communication skills, and confidence.

The major innovation of this project is the Personalized AI Tutor. Unlike existing systems that only generate questions and display scores, the AI Tutor acts as a virtual mentor by explaining incorrect answers, identifying weak topics, recommending learning resources, and generating a personalized learning roadmap. The system also adapts the difficulty level of interview questions according to the user's performance and supports company-specific interview preparation for organizations such as TCS, Infosys, Accenture, Capgemini, Deloitte, and Cognizant. A performance dashboard is included to monitor interview scores, topic-wise improvement, and overall progress.

The scope of the project includes AI-based mock interviews, resume analysis, coding assessments, personalized tutoring, adaptive learning, and performance analytics. It is intended to support candidates throughout their interview preparation journey but does not replace real interviewers or guarantee job placement. The expected outcome of the project is to improve candidates' technical knowledge, communication skills, confidence, and interview performance through continuous AI-driven mentoring, making the system more effective than conventional AI interview preparation platforms.

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### References:

- 1.Shahri, H., Emad, M., Ibrahim, N., Rais, R. N. B, & Al-Fayoumi, Y. (2024). Elevating Education through AI Tutor: Utilizing GPT-4 for Personalized Learning. Proceedings of the 15th Annual Undergraduate Research Conference on Applied Computing (URC 2024), IEEE. DOI: 10.1109/URC62276.2024.10604578.
- 2.Frankford, E., Sauerwein, C., Bassner, P., Krusche, S., & Breu, R. (2024). AI-Tutoring in Software Engineering Education: Experiences with Large Language Models in Programming Assessments. Proceedings of the 46th IEEE/ACM International Conference on Software Engineering (ICSE-SEET 2024). DOI: 10.1145/3639474.3640061.
- 3.Development of LLM-Based System for IT Talent Interview. (2024). 2024 IEEE International Conference on Data and Software Engineering (ICoDSE). DOI: 10.1109/ICoDSE63307.2024.10829889.
- 4.Xu, H., Gan, W., Qi, Z., Wu, J., & Yu, P. S. (2024). Large Language Models for Education: A Survey. arXiv, 2024.
- 5.Zhao, Y., et al. (2024). Advancing Generative Intelligent Tutoring Systems with GPT-4: Design, Evaluation, and a Modular Framework for Future Learning Platforms. Electronics, 13(24), 4876.

### Project Team Members

| S. No. | Regd. No.  | Name of the Student        | Signature |
|--------|------------|----------------------------|-----------|
| 1.     | 24BQ5A6137 | Reddy Vishnu Vardhan Reddy |           |
| 2.     | 23BQ1A61D4 | Syed Apsana Parveen        |           |
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| 4.     | 23BQ1A61A9 | Pinapati Kousalya          |           |
|        |            |                            |           |

### Project Guide

Name: Dr.Shaik Khader Zelani

Signature:

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### INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

#### Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the '**Project category**' section. Also provide the link to the problem statement in the '**References**' section.

#### To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

#### To write category specific references

|                             |                                                                                                                                                                                                     |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research oriented</b>    | Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant. |
| <b>Application oriented</b> | Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.     |
| <b>Product oriented</b>     | Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.                    |